MAT 3103: Computational Statistics and Probability

Chapter 2: Descriptive Statistics

Descriptive statistics:

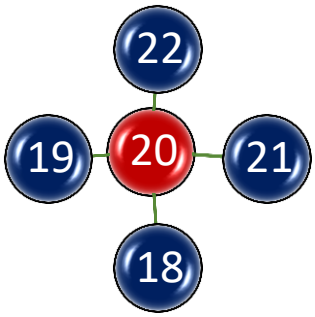
Descriptive statistics are brief descriptive measures that summarize a given data set, which can be either a representation of the entire or a sample of a population. Descriptive statistics are broken into

i) **Measures of central tendency**
ii) **Measures of dispersion**

Descriptive statistics are very important because if we simply present our raw data it would be hard to visualize what the data is showing, especially if there is a lot of it. Descriptive statistics therefore enables us to present the data in a more meaningful way, which allows simpler interpretation of the data.

Central tendency:

Central tendency is a descriptive summary of a dataset through a single value that reflects the center of the data distribution. As we can see, average of the values is 20 and all the values have a tendency to nearing 20 (center).



Measures of Central Tendency: Measures of central tendency are statistical constants which enable us to comprehend in a single effort the significance of the whole. A number which represents the entire list of numbers. A value which describes some attributes of the population. It helps us to condense data in a single value. A score that indicates where the center of the distribution tends to be located.

There are different types of measures of central tendency; each has its own advantages and disadvantages. These are -

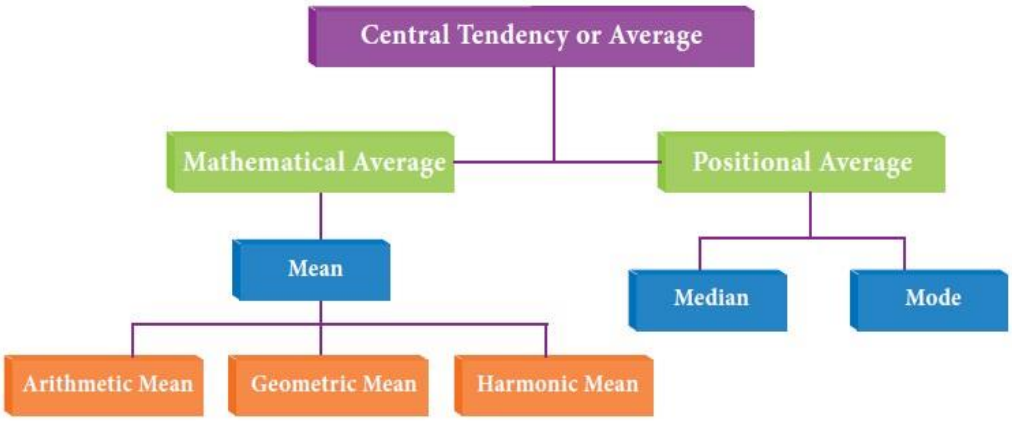
i. Mean **a) Arithmetic mean**

b) Geometric mean

c) Harmonic mean

ii. Median

iii. Mode



Source: https://www.brainkart.com/article/Various-measures-of-central-tendency 35079/

Arithmetic mean:

It is generally known as average. To find the mean, add up all the numbers and divide by the number of numbers. It is denoted as $\bar{x}$ or AM.

Calculation procedures:

For ungrouped data: Let $x_1, x_2, \ldots, x_n$ are n variates, then, the arithmetic mean is defined by

$$\text{AM} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

For grouped data: Let $x_1, x_2, \ldots, x_n$ are n variates with frequencies $f_1, f_2, f_3, \ldots, f_n$ then, the arithmetic mean is defined by

$$\text{AM} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} f_i x_i,$$

where x_i = midpoints of groups, f_i = frequency, $n = \sum_{i=1}^{n} f_i$.

Advantages of the arithmetic mean:

- It is rigidly defined.
- It is easy to calculate and simple to follow.
- It is based on all the observations.
- It is determined for almost every kind of data.

Disadvantages of the arithmetic mean:

- The arithmetic mean is highly affected by extreme values.
- It cannot average the ratios and percentages properly.
- It is not an appropriate average for highly skewed distributions.
- It cannot be computed accurately if any item is missing.

Geometric mean:

Geometric mean is relevant when several quantities multiply together to produce a product or there is geometric progression in data. It is sometimes preferred for averaging ratios of two variables: rates of population growth, rates of interest, and rates of depreciation. The geometric mean of a set of n values of a variable is the nth root of their product.

Suppose you have an investment which earns 10% the first year, 50% the second year, and 30% the third year. What is its average rate of return? It is not the arithmetic mean, because what these numbers mean is that on the first year your investment was multiplied (not added to) by 1.10, on the second year it was multiplied by 1.60, and the third year it was multiplied by 1.20. The relevant quantity is the geometric mean of these three numbers.

Calculation procedures:

For ungrouped data: Let a variable *x* assumes *n* values $x_1, x_2, \dots, x_n$. Then, geometric mean defined as,

$$\text{GM} = \bar{x}_G = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}.$$

For grouped data: Let a variable *x* assumes *n* values $x_1, x_2, \dots, x_n$ with respective frequencies as $f_1, f_2, \dots, f_n$. Then,

$$\text{GM} = \bar{x}_G = \sqrt[n]{(x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n})} = \left(\prod_{i=1}^{n} {x_i}^{f_i}\right)^{\frac{1}{n}},$$

$$\text{alternatively, } \bar{x}_G = \text{Antilog}\left(\frac{1}{n}\sum_{i=1}^{n} f_i log x_i\right),$$

where x_i = midpoints of groups, f_i = frequency, $n = \sum_{i=1}^{n} f_i$.

Advantages of the geometric mean:

- The geometric mean is rigidly defined.
- The geometric mean is directly based on all the observations.
- Extremely small or large values has no considerable effect on geometric mean.

Disadvantages of the geometric mean:

- It is difficult to compute.

- If a single value of a variable is zero, then the geometric mean becomes zero, irrespective of the magnitudes of the other values.
- It may be imaginary if some values are negative.

Harmonic mean:

Harmonic mean is helpful when dealing with datasets of rates or ratios (i.e. fractions) over different lengths or periods. For example, in first test a typist types 400 words in 50 minutes, in second test he types the same words (400) in 40 minutes and in third test he takes 30 minutes to type the 400 words. Then average time of typing can be calculated by harmonic mean.

Calculation procedures:

For ungrouped data: Let a variable *x* assumes *n* values $x_1, x_2, \dots, x_n$. Then, harmonic mean

$$\text{HM} = \bar{x}_H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}.$$

For grouped data: Let a variable *x* assumes *n* values $x_1, x_2, \dots, x_n$ with respective frequencies as $f_1, f_2, \dots, f_n$. Then, $\text{HM} = \bar{x}_H = \frac{n}{\sum_{i=1}^{n} \frac{f_i}{x_i}}$,

where x_i = midpoints of groups, f_i = frequency, $n = \sum_{i=1}^{n} f_i$.

Advantages of the harmonic mean:

- It is based on all observations.
- It is capable of algebraic treatment.
- It is an appropriate average for averaging ratios and rates.
- It does not give much weight to the large items

Disadvantages of the harmonic mean:

- It gives high weight-age to the small items.
- It cannot be calculated if any one of the items is zero.
- It is usually a value which does not exist in the given data.

Interpretation:

1. Arithmetic Mean (AM)

   This is the simple average of the quiz scores. It assumes that each score contributes equally and additively to the total. It is most commonly used when values are of the same unit and represent a total or overall level.

2. Geometric Mean (GM)

This is slightly less than the AM. The GM is appropriate when values are multiplicative in nature (e.g., growth rates, percentages), or when comparing normalized performance. In this case, it gives a central tendency that is less sensitive to extreme values than the AM.

3. Harmonic Mean (HM)

   This is the lowest of the three means. HM is most suitable when dealing with rates (e.g., speed, efficiency). It gives more weight to smaller values, which is useful when the lowest scores are particularly important in the analysis.

Example 2.1: The quiz scores of 5 randomly selected students in a section of Mathematics course at AIUB are recorded as: 15, 14, 13, 17, and 15. Calculate arithmetic mean, geometric mean and harmonic mean for the given scores of the students.

Solution: Let first arrange the values in order as: 13, 14, 15, 15, 17. Then,

Arithmetic Mean (AM): $\bar{x} = \frac{1}{5}(13 + 14 + 15 + 15 + 17) = 14.80$

Geometric Mean (GM): $\bar{x}_G = \sqrt[5]{13 \times 14 \times 15 \times 15 \times 17} = 14.74$

Harmonic Mean (HM): $\bar{x}_H = \frac{5}{\frac{1}{13}+\frac{1}{14}+\frac{1}{15}+\frac{1}{15}+\frac{1}{17}} = 14.68$

In this example, all three means are **close in value**, indicating that the data is fairly symmetric without large outliers. The slight decrease from AM to GM to HM is typical when using these three averages on positive values.

Example 2.2: A cyclist trains for three days, riding at different speeds and showing varying rates of improvement each day. Determine the most suitable methods to calculate the average speed and the overall improvement factor across the training period.

Day	Improvement Factor (%)	Speed (km/h)
Day 1	7	20
Day 2	12	26
Day 3	19	30

Solution: The improvement factor is given in percentages, which implies relative/percentage changes. The appropriate way to find the average improvement factor over multiple days is using the geometric mean, since percentage improvements are multiplicative in nature.

First, convert the percentages to factors:

Day 1: $1 + \frac{7}{100} = 1.07$

Day 2: $1 + \frac{12}{100} = 1.12$

Day 3: $1 + \frac{19}{100} = 1.17$

Now calculate the **geometric mean**:

Geometric Mean (GM): $\bar{x}_G = \sqrt[3]{1.07 \times 1.12 \times 1.19} = 1.128$

Now convert back to a percentage:

$$(1.128 - 1) \times 100 = 12.8\%$$

So, the cyclist's average improvement rate per day is approximately 12.8%**.**

The harmonic mean is used to calculate the average speed because speed involves reciprocal time calculations (i.e., total distance divided by total time).

Harmonic Mean (HM): $\bar{x}_H = \frac{3}{\frac{1}{20}+\frac{1}{26}+\frac{1}{30}} = 24.63$ km/h

So, the cyclist's average speed over the three days is 24.63 km/h**.**

Example 2.3: The distribution of number of faded signals of a day sent from different stations:

Class Interval of faded signals	1 - 3	3 - 5	5 - 7	Total
No. of stations (*f*)	2	5	3	10

Calculate arithmetic mean, geometric mean and harmonic mean for the distribution.

Solution:

Class	Frequency (*f*)	Mid value(*x*)	*fx*	$f \log x$	$\frac{f}{x}$
1 – 3	2	2	4	2log2	1
3 – 5	5	4	20	5log4	1.25
5 – 7	3	6	18	3log6	0.5
Total	n = 10		42	5.95	2.75

Arithmetic Mean (AM): $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} f_i x_i = \frac{42}{10} = 4.20$

On average, each station experiences about 4.2 faded signals in a day.

Geometric Mean (GM): $\bar{x}_G =$ Antilog $(\frac{1}{n}\sum_{i=1}^{n} f_i log x_i)$ = Antilog $(\frac{5.95}{10})$ = 3.9

This tells us that the typical (central) level of faded signals is about 3.9 per station.

Harmonic Mean (HM): $\bar{x}_H = \frac{n}{\sum_{i=1}^{n} \frac{f_i}{x_i}} = \frac{10}{2.75} = 3.63$

This value emphasizes the lower numbers (stations with fewer faded signals). The effective average number of faded signals is closer to 3.63.

Example 2.4: Show by examples that, AM ≥ GM ≥ HM.

Solution: Let a set of data *x:* 2, 3, 4, 5, 6.

AM = $\frac{2+3+4+5+6}{5} = \frac{20}{5} = 4$

GM = $\sqrt[5]{2 \times 3 \times 4 \times 5 \times 6} = \sqrt[5]{720} = 3.7279$

HM = $\frac{5}{\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}} = 3.4483$

In this case, AM > GM > HM

If data set is as *x*: 4, 4, 4, 4, 4.

Then, AM = GM = HM = 4

Note: As we can see, AM > GM > HM. If all the values are same, then AM = GM = HM. We must not use GM and HM for a dataset having a value zero (0) as no matter what the remaining values are, the result will always be 0 due to that single 0.

Example 2.5: The following is the age distribution of 1000 persons working in a large industrial house:

Age Group	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65
No. of person	30	160	210	180	145	105	70	60	40

Calculate mean for the above data.

Solution:

Age group	**No. of person (*f*)**	**Mid Value (*x*)**	***fx***
20-25	30	22.5	675
25-30	160	27.5	4400
30-35	210	32.5	6825
35-40	180	37.5	6750
40-45	145	42.5	6162.5
45-50	105	47.5	4987.5
50-55	70	52.5	3675
55-60	60	57.5	3450
60-65	40	62.5	2500
Total	1000		39425

Mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} f_i x_i = \frac{39425}{1000} = 39.425$

This value indicates the average age of all workers. Although individuals may be younger or older, the "center" of the age distribution lies around 39.43 years.

Helps HR or management understand the workforce age structure (e.g., if training or retirement planning is needed).

Median:

The median is the number that is halfway into the dataset. It overcomes the limitation of arithmetic means' inability to deal with outliers (extreme values). The middle number; found by ordering all data points and picking out the one in the middle (or if there are two middle numbers, taking the mean of those two numbers) is the median.

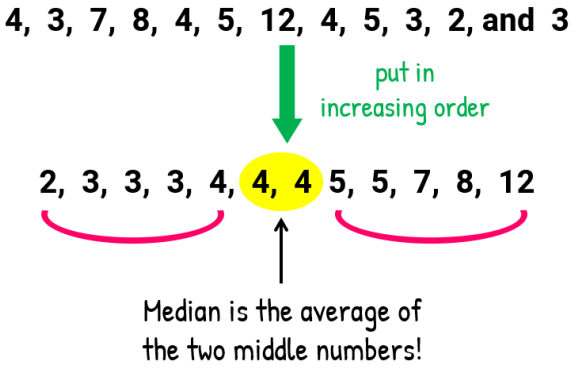


Image source: https://www.khanacademy.org/math/probability/data-distributions-a1/summarizing-center-distributions/a/choosing-the-best-measure-of-center

Calculation procedures:

Median is the middle value in the arrayed data (data arranged in either ascending or descending order).

For ungrouped data,

$$Me = \text{The value of } \frac{1}{2}(n+1)th\ observation \qquad [n\ odd]$$

$$= \text{The value of } \frac{1}{2}\left(\frac{n}{2}th\ observation + (\frac{n}{2}+1)th\ observation\right) \qquad [n\ even]$$

For grouped data,

$$\text{Median} = L + \frac{\frac{n}{2} - c}{f} \times h$$

L = lower limit of median class

h = size of median class

f = frequency of median class

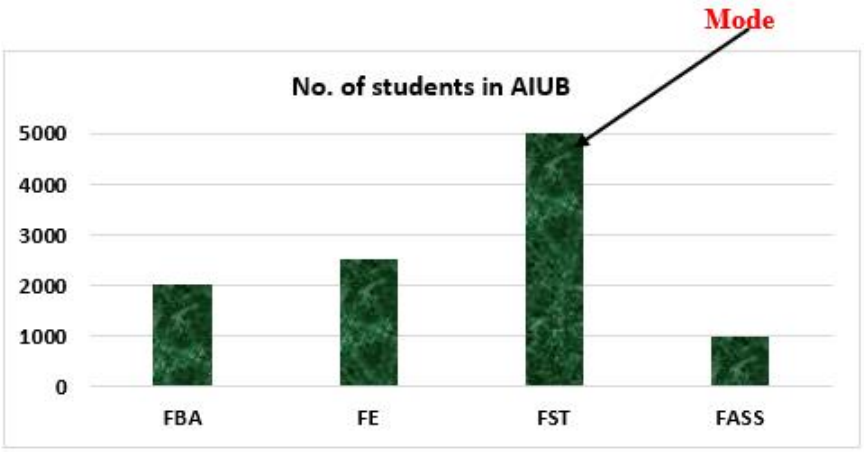
c = cumulative frequency of previous class of median class

Median class is that class for which cumulative frequency, $cf \geq \frac{n}{2}$.

Mode:

Mode is the most frequently occurring number found in a dataset. It is mainly used in situations where the variable under consideration is qualitative in nature. There can be more than one mode in a data set. If the two values are tied for being the most common values in the set, the data set can be said to be *bimodal*, whereas if three values are tied, the set is *trimodal*, and so on.

We can use bar diagram to find the mode for qualitative data.



Example 2.6:

The data set is given as x: 2, 3, 3, 4, 6. Mode = 3.

The data set is given as x: 2, 3, 3, 4, 6, 6. Mode = 3 and 6.

The data set is given as x: 2, 3, 3, 4, 4, 4, 6. Mode = 4

The data set is given as x: 2, 3, 4, 6. Mode = No mode.

For grouped data, Mode = $L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$

L = lower limit of modal class

h = size of modal class

f_m = frequency of modal class

f_1 = frequency of previous class of modal class

f_2 = frequency of next class of modal class

Modal class is that class for which frequency is the highest one.

Example 2.7: The following data give the monthly wages in taka of 7 workers of a factory. 2700, 2750, 2680, 2790, 2760, 2720, 2740. Compute median wages of the workers.

Solution: array of x: 2680, 2700, 2720, 2740, 2750, 2760, 2790.

$n = 7$ (an odd number)

$$Me = \text{The value of } \frac{1}{2}(n+1)\text{th number observation}$$

$$= \text{The value of 4th number observation}$$

$$= \text{Tk. } 2740 \text{ per month}$$

This means that half of the workers (3 workers) earn less than or equal to Tk. 2740, and the other half (3 workers) earn greater than or equal to Tk. 2740 per month.

Example 2.8: The following data refer to the profits of a store in thousand taka for the last 12 months are 3, 6, 8, 9, 6, 10, 5, 12, 9, 8, 11, 7. Compute median profit of the store.

Solution: Array of x: 3, 5, 6, 6, 7, 8, 8, 9, 9, 10, 11, 12.

$n = 12$ (an even number)

$$Me = \text{The value of } \frac{1}{2}\left(\frac{n}{2}\text{th number observation} + \left(\frac{n}{2}+1\right)\text{th number observation}\right)$$

$$= \text{The value of } \frac{1}{2}(6\text{th number observation} + 7\text{th number observation})$$

$$= \frac{8+8}{2} \quad = \text{Tk. } 8 \text{ thousand}$$

Example 2.9: The marks obtained by 7 students in an examination were 52, 89, 96, 93, 89, 92, 99. Find mode of the marks.

Solution: The values of the data set in ascending order is: 52, 89, 89, 92, 93, 96, 99.

Mode = 89, since it occurs two times in the data set.

In this case, 89 is the most common score among the 7 students.

Example 2.10: The following data give the speeds (in miles per hour) of eight cars that were stopped on I-95 speed violations.

77, 82, 74, 81, 79, 84, 74, 78, 82

Find the mode.

Solution: The values of the data set in ascending order is: 74, 74, 77, 79, 81, 82, 82, 84

Since 74 and 82 occur twice, each of the remaining values occurs only once. So,

Mode = 74 and 82 miles per hour.

Mode = 74 and 82 miles per hour, indicating a bimodal distribution with two equally frequent speeds.

Example 2.11: A café records the number of times different coffee types are ordered in a day: Latte (15), Espresso (22), Cappuccino (18), Americano (22), Mocha (12). Find appropriate measure of central tendency.

Solution: The mode is the coffee type ordered the most times. Both Espresso and Americano were ordered 22 times, which is the highest frequency.

Mode = Espresso and Americano (Bimodal Data Set)

This means that these two coffee types are the most popular in the café.

Example 2.12: The following frequency distribution refers to the data for marks obtained by students in a paper. Compute mean, median of the frequency distribution.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of students	5	7	8	10	10	10	4	4	2

Solution: The calculation is shown in the table below:

Marks	Frequency, f	Mid Value, x	fx	Cumulative frequency, cf
0-10	5	5	25	5
10-20	7	15	105	12
20-30	8	25	200	20
30-40	10	35	350	30
40-50	10	45	450	40
50-60	10	55	550	50
60-70	4	65	260	54
70-80	4	75	300	58
80-90	2	85	170	60
Total	$n = 60$		$fx = 2410$	

Mean, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} f_i x_i = \frac{2410}{60} = 40.16$

Here, $\frac{n}{2} = \frac{60}{2} = 30$, for the class $30-40$, $cf = 30 = 30$. So, median class is $30-40$.

$$Me = L + \frac{\frac{n}{2} - c}{f} \times h = 30 + \frac{\frac{60}{2} - 20}{10} \times 10 = 40$$

Example 2.13: The number of faded signals of a day sent from different stations is as:

Class Interval of faded signals	0 - 5	5 - 10	10 - 15	Total
No. of stations (*f*)	4	6	5	15

Calculate median and mode for the distribution.

Solution:

Class	*f*	*cf*
0 – 5	4	4
5 – 10	6	10
10 – 15	5	15
Total	$n = 15$	

Here, $\frac{n}{2} = \frac{15}{2} = 7.5$. For the class (5 – 10), we get $cf = 10 > 7.5$. Hence, this is our median class.

Median, $Me = L + \frac{\frac{n}{2} - c}{f} \times h = 5 + \frac{7.5 - 4}{6} \times 5 = 7.92$

For the class (5 – 10), we have the highest frequency 6. Hence, this is our modal class.

Mode, $Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h = 5 + \frac{6 - 4}{12 - 4 - 5} \times 5 = 8.33$

Example 2.14: The following frequency distribution refers to the number of hours worked per month of 50 workers of a factory. Compute median and mode of the frequency distribution.

Number of hours worked	30-55	55-80	80-105	105-130	130-155	155-180	180-205
Number of workers	3	4	6	9	12	11	5

Solution: The calculation is shown in the table below:

Number of hours worked	Frequency, *f*	Cumulative frequency, *cf*

30-55	3	3
55-80	4	7
80-105	6	13
105-130	9	22
130-155	12	34
155-180	11	45
180-205	5	50
Total	n = 50	

Here, $\frac{n}{2} = \frac{50}{2} = 25$, for the class 130-155, $cf = 34 > 25$. So, median class is 130-155.

$$Me = L + \frac{\frac{n}{2} - c}{f} \times h = 130 + \frac{\frac{50}{2} - 22}{12} \times 25 = 136.26$$

Median work hours are 136.26 hour represent the middle value, indicating that half of the workers work less than 136.26 hour, while the other half work more**.**

The class 130-155 contains the highest frequency. Hence the modal class is 130-155.

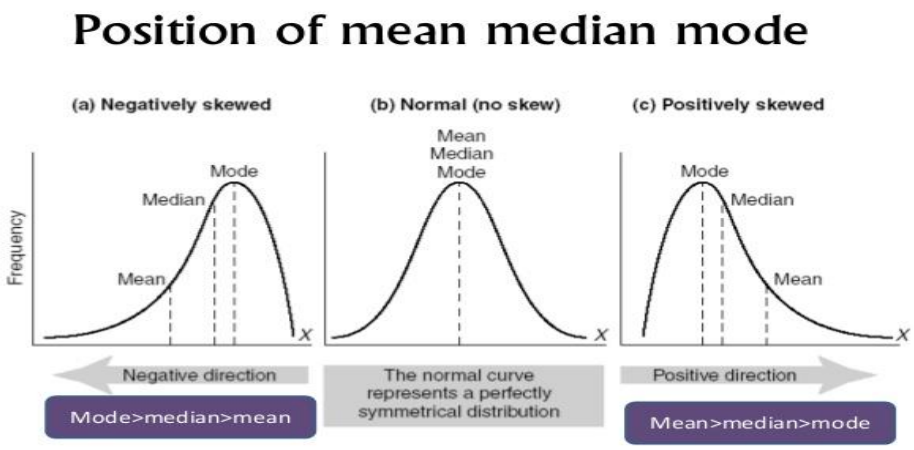
$$Mode, Mo = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h = 130 + \frac{12 - 9}{2 \times 12 - 9 - 11} \times 25 = 148.75$$

Mode work hours are 148.75 hour is the most common working hours level, meaning most workers work around 148.75 hours.

Note: If $f_m$is there in the first class, then f_1 = 0. If $f_m$is there in the last class, then f_2 = 0. If there are two or more f_m, then separate modes should be calculated considering separatef_m.

Application of central tendency measures in describing the shape of distributions:

Central tendency measures can help us in describing the shape of distributions through a measure called **skewness**. It measures the lack of symmetry in data distribution.



A **symmetrical** distribution will have a skewness of zero. If data is **positively skewed**, then it will have a much longer right tail than the left tail. If data is **negatively skewed**, then it will have a much longer left tail than the right tail.

Calculation and interpretation rules for skewness:

Measure of skewness	Interpretation rule
Mean = Median = Mode	the distribution is symmetric
Mean > Median > Mode	the distribution is positively Skewed
Mode > Median > Mean	the distribution is negatively Skewed
SK = mean – median Or SK = mean – mode	If SK = 0, the distribution would be symmetrical If SK > 0, the distribution would be positively skewed If SK < 0, the distribution would be negatively skewed

Comment on the skewness of the given distributions:

1. 4, 6, 9, 12, 5; mean = 7.2; median = 6; mode = no mode
2. 7, 13, 4, 7; mean = 7.75; median = 7; mode = 7
3. 10, 3, 8, 15; mean = 9; median = 9; mode = no mode
4. 9, 9, 9, 9, 8; mean = 8.8; median = 9; mode = 9
5. 300, 24, 40, 50, 60; mean = 96.8; median = 50; mode = no mode
6. 23, 23, 12, 12; mean = 17.5; median = 17.5; mode = 12, 23

Dispersion:

Dispersion is a way of describing how a set of data is spread. When a data set has a large value, the values in the set are widely scattered; when it is small the items in the set are tightly clustered.

Measures of central tendency might not always be helpful to describe data. Two datasets having same average could be entirely different in pattern. Let us illustrate it by a practical example. The average midterm scores of 2 courses of 5 students of CSE in AIUB are given as:

Courses	Scores					Average
Math	46	48	50	52	54	50
Statistics	10	40	50	60	90	50

In both courses, average scores are equal. But in Math, the observations are concentrated on the center. All students have almost the same level of performance. We say that there is consistence in the observations. In Statistics, the observations are not closed to the center. One observation is as small as 10 and one observation is as large as 90. Thus, there is greater dispersion in Statistics.

Measures of dispersion:

Measures of variability help communicate this by describing the shape and spread of the data set. Mean deviation, Variance, Standard deviation, coefficient of variation and quartiles are all examples of measures of variability.

Range:

Range is the simplest method of studying variation. It is defined as the difference between the value of the smallest observation and the value of the largest observation included in the distribution.

$$Range = Largest\ Value - Smallest\ Value$$

Mean deviation:

Mean deviation is the average distance between each observed value and the mean. Mean deviation is used frequently by engineers to show the variability of their data, although it is

usually not the best choice. Its advantage is that it is simpler to calculate than other measures of dispersion.

Calculation procedures:

For ungroup data: Let a variable *x* assumes values $x_1, x_2, \dots, x_n$. Then,

$$\text{Mean deviation, MD} = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$$

For grouped data: Let a variable *x* assumes values $x_1, x_2, \dots, x_n$ with respective frequencies as $f_1, f_2, \dots, f_n$. Then,

$$\text{Mean deviation, MD} = \frac{1}{n}\sum_{i=1}^{n} f_i|x_i - \bar{x}|$$

Variance:

Variance is the mean of the squares of the deviations of each observations from their mean. Note that variance has units of the quantity squared, for example m^2 or s^2 if the original quantity was measured in meters or seconds, respectively. Standard deviation overcomes this problem.

Calculation procedures:

For ungroup data: Let a variable *x* assumes values $x_1, x_2, \dots, x_n$. Then,

$$\text{Variance, } \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

For grouped data: Let a variable *x* assumes values $x_1, x_2, \dots, x_n$ with respective frequencies as $f_1, f_2, \dots, f_n$. Then,

$$\text{Variance, } \sigma^2 = \frac{1}{n}\sum_{i=1}^{n} f_i(x_i - \bar{x})^2$$

Standard deviation:

Standard deviation is the positive square root of variance. Thus, it has the same units as the original data and is a representative of the deviations from the mean. Since the variance is the mean square of the deviations from the mean, the standard deviation is the root-mean-square deviation from the mean.

$$\text{Standard deviation (SD), } \sigma = \sqrt{\text{Variance}}$$

Root-mean-square quantities are crucial in describing the alternating current of electricity. An analogy can be drawn between the standard deviation and the radius of gyration encountered in applied mechanics.

Coefficient of variation:

Coefficient of variation is the ratio between the standard deviation and the mean for the same set of data, expressed as a percentage. When its value is 20%, it means that the observations vary, on an average, 20% with respect to mean. It is a unit free measurement, used to compare different sets of data having different units of measurement.

$$\text{Coefficient of variation, CV} = \frac{\text{Standard deviation}}{Mean} \times 100\% = \frac{\sigma}{\bar{x}} \times 100\%$$

Example 2.15: The following are the prices of a company from Sunday to Thursday:

Day	Sunday	Monday	Tuesday	Wednesday	Thursday
Price (Tk)	200	210	208	160	250

Find the range.

Solution: Here,

Largest Value = 250

Smallest Value = 160

$$\begin{aligned} Range &= Largest\ Value - Smallest\ Value \\ &= 250 - 160 \\ &= 90 \text{ Tk} \end{aligned}$$

A range of Tk 90 means the company's price fluctuated up to Tk 90 between the lowest and highest prices in the week.

Example 2.16: The following data relate to the marks obtained by nine students in a class test. Compute the mean deviation from this data set.

3, 4, 4, 5, 5, 7, 7, 7, 12

Solution: $Mean,\ \bar{x} = 6$

Mean deviation, $MD = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}| = \frac{|3-6|+|4-6|+\ldots+|12-6|}{8} = 2$

The mean deviation of 2 means that, on average, the marks deviate by 2 points from the mean score of 6.

Example 2.17: The following frequency distribution gives the pattern of overtime work per week by 100 employees of a company. Calculate mean deviation of the following distribution:

Overtime (in hour)	10-15	15-20	20-25	25-30	30-35
Number of employees	3	5	7	4	2

Solution:

Overtime	x_i	f_i	$x_i f_i$	$\|x_i - \bar{x}\|$	$f_i \| x_i - \bar{x}\|$
10-15	12.5	3	37.5	9.286	27.858
15-20	17.5	5	87.5	4.286	21.43
20-25	22.5	7	157.5	0.714	4.998
25-30	27.5	4	110	5.714	22.856
30-35	32.5	2	65	10.714	21.428
Total		21	457.5		98.57

$$\bar{x} = \frac{\sum f_i x_i}{n} = \frac{457.5}{21} = 21.786$$

$$MD = \frac{1}{n}\sum_{i=1}^{n} f_i|x_i - \bar{x}| = \frac{98.57}{21} = 4.69$$

The mean deviation of 4.69 hours tells us that, on average, employees' weekly overtime deviates about 4.7 hours from the mean.

Example 2.18: A company of tea prices at 6 randomly selected grocery stores in an area of Sylhet city showed increase of 2, 4, 8, 6, 10, 12 Tk. per kilogram from the previous month. Find the variance, standard deviation and coefficient of variation of the price increases.

Solution:

x_i	$(x_i - \bar{x})^2$
2	$(2-7)^2 = 25$
4	$(4-7)^2 = 9$
8	$(8-7)^2 = 1$
6	$(6-7)^2 = 1$
10	$(10-7)^2 = 9$
12	$(12-7)^2 = 25$
$\sum x_i = 42$	$\sum(x_i - \bar{x})^2 = 70$

$$\bar{x} = \frac{\sum x}{n} = \frac{42}{6} = 7$$

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{n} = \frac{70}{6} = 11.67$$

$$\sigma = \sqrt{11.67} = 3.42$$

$$CV = \frac{\sigma}{\bar{x}} \times 100 = 48.86\%$$

The standard deviation of Tk. 3.42 indicates that price increases typically vary about 3.42 Tk. from the mean.

The coefficient of variation (CV = 48.86%) shows moderate-to-high relative variability in price increases across stores. A higher CV suggests less consistency, prices are increasing at different rates among the stores.

Example 2.19: The following frequency distribution is the data of days to maturity 40 short term investment. Calculate standard deviation and coefficient of variation.

Class Interval	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	3	1	8	10	7	7	4

Solution:

Class Interval	f_i	x_i	$f_i\, x_i$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$
30-40	3	35	105	1122.25	3366.75
40-50	1	45	45	552.25	552.25
50-60	8	55	440	182.25	1458
60-70	10	65	650	12.25	122.5
70-80	7	75	525	42.25	295.75
80-90	7	85	595	272.25	1905.75
90-100	4	95	380	702.25	2809
Total	40		2740		10510

$\bar{x} = \frac{\sum f_i x_i}{n} = \frac{2740}{40} = 68.5 \approx 67$

$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{n} = \frac{10510}{40} = 262.75$

$\sigma = \sqrt{262.75} = 16.21$

Coefficient of Variation, $CV = \frac{\sigma}{\bar{x}} \times 100 = 23.66\%$

The average time to maturity of the short-term investments is 68.5 days.

The standard deviation of 16.21 days indicates that the days to maturity typically vary by about 16 days from the mean.

The coefficient of variation (23.66%) reflects the relative variability. Since it's below 25%, the data shows moderate consistency in maturity periods.

Example 2.20: The quiz scores of a student in a Math course at AIUB are recorded as: 10, 0, and 20. Calculate mean deviation, variance, standard deviation and coefficient of variation for the given scores of that student.

Solution:

Quiz	x	$\bar{x}$	x_i - $\bar{x}$	$\lvert x_i - \bar{x}\rvert$	$(x_i - \bar{x})^2$
1	10	30/3 = 10	10 – 10 = 0	0	0
2	0		0 – 1 0= -10	10	100
3	20		20 – 10= 10	10	100
Total	30		0	20	200
MD $= \frac{1}{n}\sum_{i=1}^{n}\lvert x_i - \bar{x}\rvert = \frac{20}{3} = 6.67$				$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{200}{3} = 66.67$	
SD $= \sigma = \sqrt{\text{Variance}} = \sqrt{66.67} = 8.16$				CV $= \frac{\mathbf{8.16}}{\mathbf{1}} \times 100\% = 81.65\%$	

Example 2.21: Track the weekly internet data usage (in GB) at a startup office to measure consistency and variation in usage over several weeks.

Class Interval of (Internet Usage in GB)	50 - 60	60 - 70	70- 80	80 - 90	Total
No. of Weeks (*f*)	4	7	6	3	20

Calculate mean deviation, variance, standard deviation and coefficient of variance for the distribution.

Solution:

Class	f	x	fx	$\bar{x}$	x_i - $\bar{x}$	$f_i\lvert x_i - \bar{x}\rvert$	$f_i(x_i - \bar{x})^2$
50 – 60	4	55	280	1380/20 = 69 GB	55 – 69 = -14	56	784
60 – 70	7	65	455		65 – 69 = -4	28	112
70 – 80	6	75	450		75 – 69 = 6	36	216
80 – 90	3	85	255		85 – 69 = 16	48	768
Total	n = 20		1380			168	1880
Mean deviation, MD $= \frac{1}{n}\sum_{i=1}^{n} f_i\lvert x_i - \bar{x}\rvert = \frac{168}{20}$ = 8.4 GB					Variance, $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n} f_i(x_i - \bar{x})^2 = \frac{1880}{20}$ = 94 GB^2		
Standard deviation, $\sigma = \sqrt{\text{Variance}} = \sqrt{94}$					Coefficient of variation, CV $= \frac{\mathbf{9.7}}{\mathbf{69}} \times 100\%$		

= 9.7 GB	= 14.06%

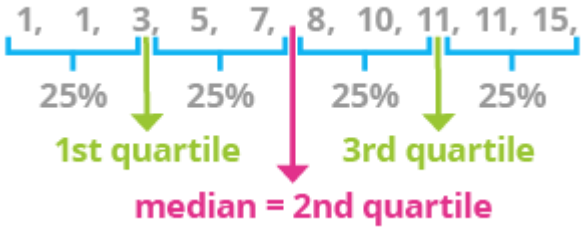
Quartiles:

Quartiles are the values that break down the dataset into quarters, or quartiles:

The first quartile (Q_1) is the point below which a quarter of the data lies. It is sometimes called the lower quartile.

The second quartile (Q_2) is the point below which half of the data lies. We already know this point by the name of median. It is also called the middle quartile.

The third quartile (Q_3) is the point below which three-quarters of the data lies. It is called the upper quartile.



Application of quartiles:

Quartiles are used to summarize a group of numbers. Instead of looking a big list of numbers, you are looking at just a few numbers that give you a picture of what's going on in the big list. Quartiles are great for reporting on a set of data and for making box and whisker plots. Quartiles are good for NOT symmetrically distributed data set, or a data set that has outliers.

Calculation procedure:

For ungrouped data: Let a variable *x* takes *n* values $\boldsymbol{x_1, x_2, \dots, x_n}$. After arranging the observations in an ascending order, the quartiles are evaluated as:

$$Q_i = The\ value\ of \frac{i(n+1)}{4} th\ observation \qquad [n\ odd];\ \ i = 1,2,3$$

$$= The\ value\ of \frac{1}{2}\left[\frac{in}{4} th\ observation + (\frac{in}{4}+1)\ th\ observation\right] \qquad [n\ even];\ \ i = 1,2,3$$

For grouped data: Quartiles, $Q_i = L + \frac{\frac{in}{4} - c}{f} \times h$; i = 1, 2, 3

L = lower limit of quartile class

h = size of quartile class

f = frequency of quartile class

c = cumulative frequency of previous class of quartile class

Quartile class is that class for which cumulative frequency, $cf \geq \frac{in}{4}$; $i = 1, 2, 3$

Example 2.22: A data set of the no. of OPD patients in a clinic in a week is given as,

x: 2, 5, 3, 6, 7, 4, 9

Calculate the value of Q_1, Q_2 and Q_3.

Solution: Make an array, x: 2, 3, 4, 5, 6, 7, 9

Here, n = 7 (odd)

$$Q_i = The\ value\ of \frac{i(n+1)}{4} th\ observation$$

$$Q_1 = The\ value\ of \frac{(n+1)}{4} th\ observation$$

$$= The\ value\ of\ 2nd\ observation$$

$$= 3$$

$$Q_2 = The\ value\ of \frac{2(n+1)}{4} th\ observation$$

$$= The\ value\ of\ 4th\ observation$$

$$= 5$$

$$Q_3 = The\ value\ of \frac{3(n+1)}{4} th\ observation$$

$$= The\ value\ of 6th\ observation$$

$$= 7$$

Example 2.23: A data set of the no. of cancer patients in a hospital is given as.

x: 2, 5, 3, 6, 7, 4, 9,13

Calculate the value of Q_1, Q_2 and Q_3.

Solution: Make an array, x: 2, 3, 4, 5, 6, 7, 9, 13

Here, n = 8 (even)

$$Q_i = The\ value\ of \tfrac{1}{2}\left[\tfrac{in}{4} th\ observation + (\tfrac{in}{4} + 1)\ th\ observation\right]$$

$$Q_1 = The\ value\ of \frac{1}{2}\left[\frac{n}{4} th\ observation + (\frac{n}{4} + 1)\ th\ observation\right]$$

$$= The\ value\ of \tfrac{1}{2}[2nd\ observation + 3rd\ th\ observation]$$

$$= \frac{1}{2}(3 + 4)$$

$$= 3.5$$

$$Q_2 = The\ value\ of\ \frac{1}{2}\left[\frac{2n}{4}th\ observation + (\frac{2n}{4}+1)\ th\ observation\right]$$

$$= The\ value\ of\ \frac{1}{2}[4th\ observation + 5th\ th\ observation]$$

$$= \frac{1}{2}(5+6)$$

$$= 5.5$$

$$Q_3 = The\ value\ of\ \frac{1}{2}\left[\frac{3n}{4}th\ observation + (\frac{3n}{4}+1)\ th\ observation\right]$$

$$= The\ value\ of\ \frac{1}{2}[6th\ observation + 7th\ th\ observation]$$

$$= \frac{1}{2}(7+9)$$

$$= 8$$

Q_1 = 3.5: 25% of patients had counts ≤ **3.5**.

Q_2 = 5.5: This is the **median**; 50% of values are ≤ **5.5**.

Q_3 = 8: 75% of values are ≤ **8**.

Example 2.24: The distribution of temperature of 20 cities is given below. Find the value of Q_1, Q_2 and Q_3 from these given data.

Temperature	No of cities	Cumulative frequency
0-10	2	2
10-20	3	5
20-30	5	10
30-40	2	12
40-50	6	18
50-60	2	20
Total	20	

Solution: $Q_i = L + \frac{\frac{in}{4} - c}{f} \times h$

For $i = 1$, $\frac{n}{4} = \frac{20}{4} = 5$. In the class (10 – 20), *cf* = 5. Hence, this is our class for Q_1.

$$Q_1 = L + \frac{\frac{n}{4} - \mathrm{c}}{f} \times h = 10 + \frac{5-2}{3} \times 10 = 20$$

For $i = 2$, $\frac{2n}{4} = \frac{40}{4} = 10$. In the class (20 – 30), *cf* = 10. Hence, this is our class for Q_2.

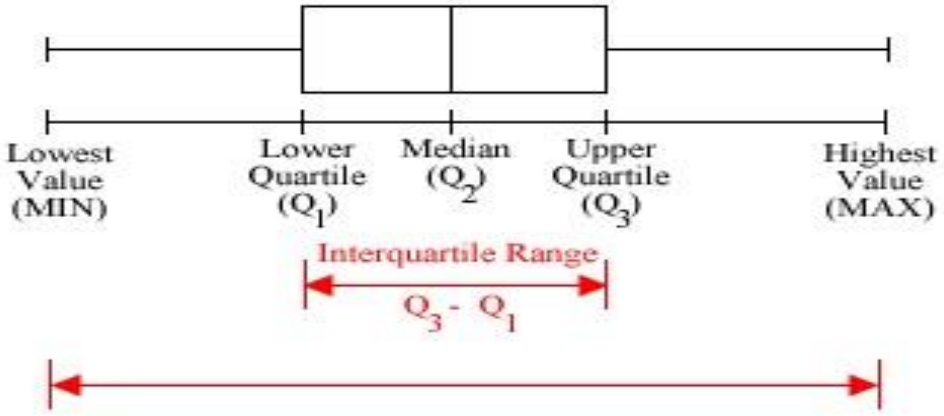
$$Q_2 = L + \frac{\frac{2n}{4} - \mathrm{c}}{f} \times h = 20 + \frac{10-5}{3} \times 10 = 36.67$$

For $i = 3$, $\frac{3n}{4} = \frac{60}{4} = 15$. In the class (40 – 50), *cf* = 18 > 15. Hence, this is our class for Q_3.

$$Q_3 = L + \frac{\frac{3n}{4} - c}{f} \times h = 40 + \frac{15 - 12}{6} \times 10 = 45$$

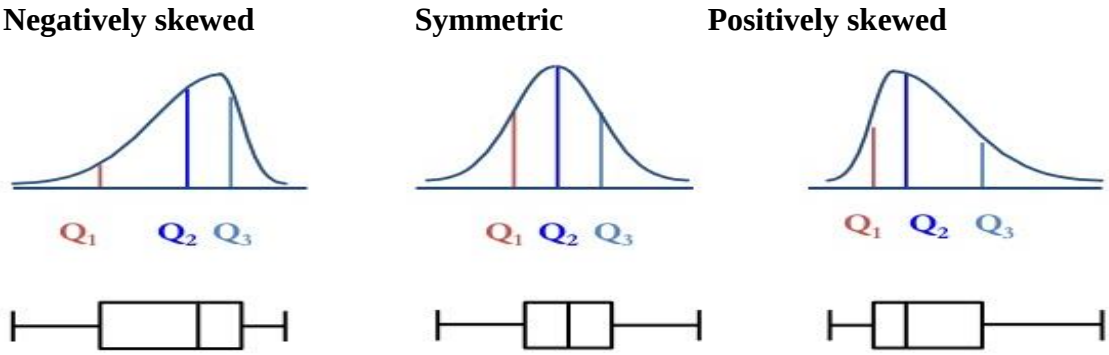
Box-and-whisker plot:

Boxplots are a standardized way of displaying the distribution of data based on a five-number summary (minimum, first quartile (Q_1), median, third quartile (Q_3), and maximum).



Application of box-and-whisker plot:

Box plots are useful as they show outliers within a data set. Box plots also show the average score of a data set. The median is the average value from a set of data and is shown by the line that divides the box into two parts. Half the scores are greater than or equal to this value and half are less. The box plot shape will show if a statistical data set is normally distributed or skewed.



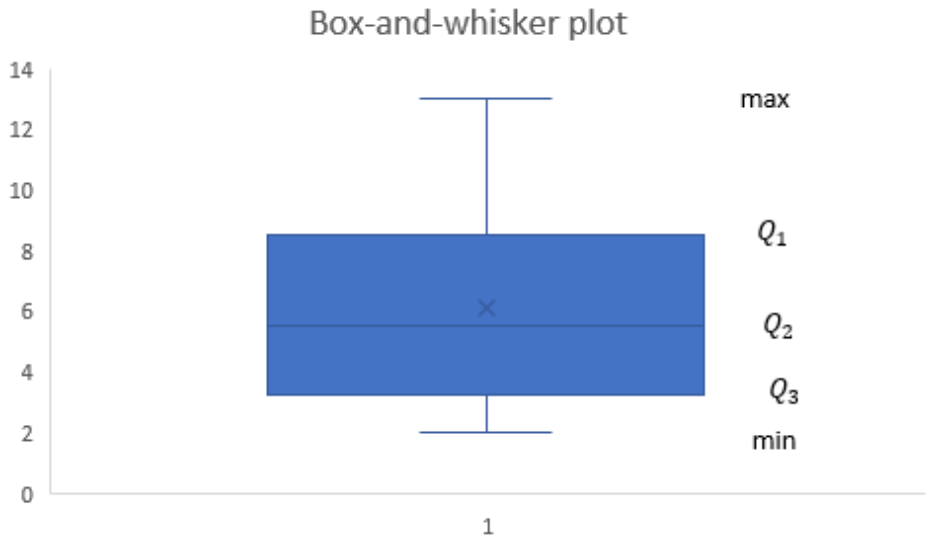
Example 2.25: A data set of the no. of cancer patients in a hospital is given as.

x: 2, 5, 3, 6, 7, 4, 9,13

Draw a box-and-whisker plot.

Solution: In the given data set (from Example 2.18)

Min = 2, Q_1 = 3.5, Q_2 = 5.5, Q_3 = 8, Max = 13.



1. Minimum (2): The lowest number of cancer patients observed.
2. Q_1 (3.5): 25% of the observations lie below or equal to 3.5 patients.
3. Median (Q_2 = 5.5): This is the middle value — 50% of the data fall below this number.
4. Q_3 (8): 75% of the data are less than or equal to 8 patients.
5. Maximum (13): The highest number of cancer patients recorded.

Screening of the Data: Data screening should be carried out prior to any statistical procedure. The screening of the data after collection and before analysis is probably the most important part of data analysis. Often data screening procedures are so tedious that they are skipped. Then, after an analysis produces unanticipated results, the data are scrutinized. This step is, however, of utmost importance as it provides the foundation for any subsequent analysis and decision-making which rests on the accuracy of the data. This procedure performs a screening of data in a database, reporting on the:

1. Detect and correct data errors
2. Type of data (discrete or continuous)
3. Missing-value patterns

4. Data transformations & standardizations
5. Presence of outliers

Detect and correct data errors: Examine summary statistics (e.g., n, mean, min, max) and check for irregularities.

Where did all the data go?

	variable	nobs	min	max	mean	median	sum	sd	cv	zeros	pct.zeros	missing	pct.missing	se	se.ratio
1	AMGO	32	1	1	1.000	1	1	NA	NA	0	0.000	31	96.875	NA	NA
2	AMRO	32	0	5	0.625	0	20	1.129	180.640	21	65.625	0	0.000	0.200	32.000
3	BCCH	32	0	2	0.125	0	4	0.421	336.800	29	90.625	0	0.000	0.074	59.200
4	BEKI	32	0	0	0.000	0	0	0.000	NaN	32	100.000	0	0.000	0.000	NaN
5	BEWR	32	0	1	0.062	0	2	0.246	396.774	30	93.750	0	0.000	0.043	69.355
6	BGWA	32	0	2	1.379	2	40	0.942	68.310	9	28.125	3	9.375	0.175	12.690
7	BHGR	32	1	250	10.875	5	348	43.680	401.655	0	0.000	0	0.000	7.722	71.007

Unrealistic value?

Action:

- Correct errors in the raw data.
- Fixing or removing incorrect data.
- Identify unusual cases.
- Distinguish duplicate cases.
- Manual check for other anomalies.

Missing-value patterns: Evaluate volume and pattern of missing data and take corrective action, if needed:

e.g., median replacement

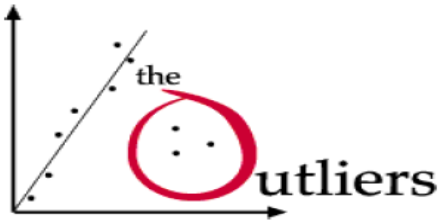
	AMGO	AMRO	BCCH	BEKI	BEWR	BGWA	BHGR
1	0	1	0	5	0	NA	25
2	NA	0	0	0	0	0	4
3	0	5	2	0	0	2	1
4	0	0	0	0	0	2	1
5	0	3	0	0	1	2	1
6	1	1	0	0	0	2	1
7	0	0	0	0	0	2	1
8	0	0	0	0	0	2	1
9	0	0	0	0	0	2	1
10	0	0	0	0	0	2	1
11	0	1	1	0	0	2	1
12	0	2	0	0	0	2	1
13	0	0	1	0	0	0	5
14	0	2	0	0	1	0	5
15	0	1	0	0	0	NA	5
16	0	1	0	0	0	NA	5

	AMGO	AMRO	BCCH	BEKI	BEWR	BGWA	BHGR
1	0	1	0	5	0	2	25
2	0	0	0	0	0	0	4
3	0	5	2	0	0	2	1
4	0	0	0	0	0	2	1
5	0	3	0	0	1	2	1
6	1	1	0	0	0	2	1
7	0	0	0	0	0	2	1
8	0	0	0	0	0	2	1
9	0	0	0	0	0	2	1
10	0	0	0	0	0	2	1
11	0	1	1	0	0	2	1
12	0	2	0	0	0	2	1
13	0	0	1	0	0	0	5
14	0	2	0	0	1	0	5
15	0	1	0	0	0	2	5
16	0	1	0	0	0	2	5

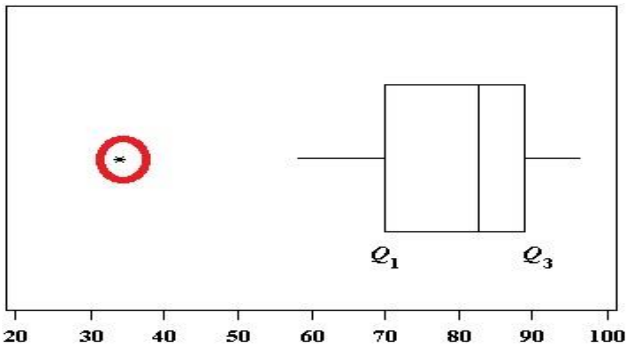
Action: Replace with prior knowledge; insert means or medians; use regression to estimate values.

Finding outliers:

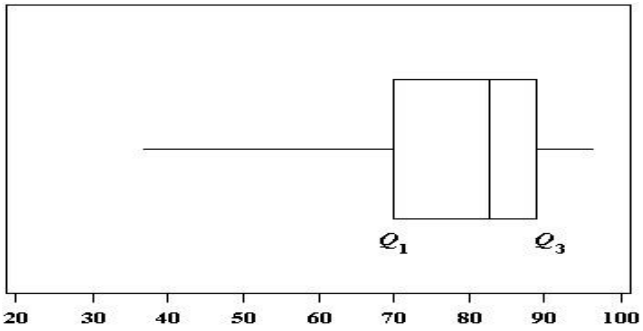
To ease the discovery of outliers, we have plenty of methods in statistics, but we will only be discussing two of them. Mostly we will try to see visualization methods (easiest ones) rather mathematical.



An outlier is an observation that is numerically distant from the rest of the data, that is, it lies outside the other values in the set. Box plots are useful as they show outliers within a data set. The whiskers of a box-and-whisker chart reach out to include outliers.



Some boxplots may not show outliers. For example, this chart has whiskers that reach out to include outliers:



Therefore, don't rely on finding outliers from a box and whiskers chart. That said, box and whiskers charts can be a useful tool to display them after calculating what the outliers are.

The most effective way to find all of your outliers is by using the interquartile range (IQR). The IQR contains the middle bulk of the data.

If a data point is below **Low value** or above **High value**, it will be defined as an outlier.

Low = Q_1– 1.5* IQR

High = Q_3+ 1.5* IQR

IQR $= Q_3 - Q_1$

Example 2.26: For the given data set identify is there any outlier if there find the value and remove this.

x: 30, 16, 70, 29, 22, 19, 3, 32, 40, 12

Solution:

$Q_1 = 14$

$Q_2 = 25.5$

$Q_3 = 36$

IQR = 22

High = 36 + 1.5*22 = 69

Low = 14 – 1.5*22 = – 19

The data set in order

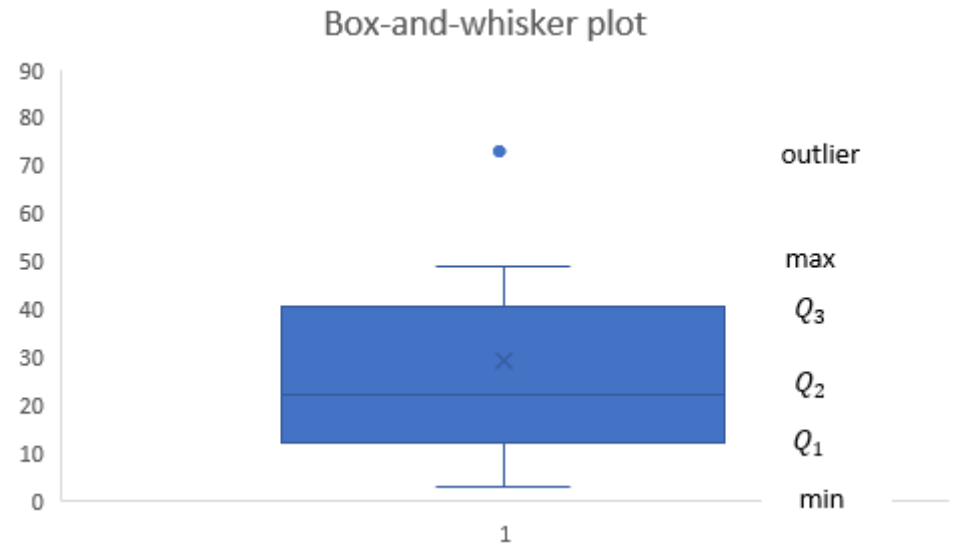
3, 12, 16, 19, 22, 29, 30, 32, 40, 70

Insert low and high values into the data set, in order

-19, 3, 12, 16, 19, 22, 29, 30, 32, 40, **69**, 70

So, outlier is 70. For further analysis remove the value.

Visualization of outlier by box-and-whisker plot.



So, bad data, wrong calculation, these can be identified as outliers and should be dropped but at the same time you might want to correct them too, as they change the level of data i.e. mean which cause issues when we model our data. For example 5 people get salary of 10K, 20K, 30K, 40K and 50K and suddenly one of the person start getting salary of 100K. Consider this situation as, you are the employer, the new salary update might be seen as biased and you might need to increase other employee's salary too, to keep the balance. So, there can be multiple reasons you want to understand and correct the outliers.

SPSS code

1. Descriptive Statistics (Mean, Median, Standard Deviation, etc.)

    - Step 1: Go to Analyze > Descriptive Statistics > Descriptives.
    - Step 2: Select the variables you want to analyze and move them to the Variables box.
    - Step 3: Click on the Options button and choose the statistics you need (mean, standard deviation, range, etc.).
    - Step 4: Click OK to generate the output.

2. Frequency Distribution (for Count and Percentage)

    - Step 1: Go to Analyze > Descriptive Statistics > Frequencies.
    - Step 2: Select the categorical variables and move them to the Variables box.
    - Step 3: Click on Statistics if you need additional options like the mode, median, or percentiles.
    - Step 4: Click on Charts to choose a chart type (like bar chart or pie chart) for visual representation.
    - Step 5: Click OK to generate the output.

3. Measures of Central Tendency (Mean, Median, Mode)

    - Step 1: Go to Analyze > Descriptive Statistics > Descriptives.
    - Step 2: Select the variables you want to analyze and move them to the Variables box.
    - Step 3: Click on the Options button.
    - Step 4: Check the options for Mean, Median, and Mode.

- Step 5: Click OK to generate the output.

4. Measures of Dispersion (Range, Variance, Standard Deviation)

- Step 1: Go to Analyze > Descriptive Statistics > Descriptives.
- Step 2: Select your variables and move them to the Variables box.
- Step 3: Click on the Options button.
- Step 4: Check Standard deviation, Range, and Variance.
- Step 5: Click OK to generate the output.

5. Skewness and Kurtosis (for Distribution Shape)

- Step 1: Go to Analyze > Descriptive Statistics > Descriptives.
- Step 2: Select your variables and move them to the Variables box.
- Step 3: Click on Options.
- Step 4: Check the boxes for Skewness and Kurtosis.
- Step 5: Click OK to generate the output.

6. Boxplot (for Visualizing Data Spread and Outliers)

- Step 1: Go to Graphs > Legacy Dialogs > Boxplot.
- Step 2: Choose the type of boxplot you want (e.g., simple, clustered).
- Step 3: Select the variables for the boxplot.
- Step 4: Click OK to generate the boxplot.

7. Explore (for Summary Statistics and Plots)

- Step 1: Go to Analyze > Descriptive Statistics > Explore.
- Step 2: Move the variables you want to explore into the Dependent List box.
- Step 3: Under Statistics, select Descriptives and Percentiles if needed.
- Step 4: Under Plots, choose Boxplot or Stem-and-leaf if you want a visual representation.
- Step 5: Click OK to generate the output.

8. Crosstabs (for Bivariate Statistics between Two Variables)

- Step 1: Go to Analyze > Descriptive Statistics > Crosstabs.
- Step 2: Select the row and column variables for the cross-tabulation.
- Step 3: Click on Statistics to select options like Chi-Square if you need significance tests.
- Step 4: Click OK to generate the crosstab table.

9. Histogram (for Visual Distribution of Data)

- Step 1: Go to Graphs > Legacy Dialogs > Histogram.
- Step 2: Select the variable you want to visualize.
- Step 3: Choose Display normal curve if you want to add a normal distribution curve.
- Step 4: Click OK to generate the histogram.

Exercise 2

1. Show, by examples, that AM ≥ GM ≥ HM.
2. The mean weight of three dogs is 38 pounds. One of the dogs weight is 46 pounds. The other two dogs, Eddie and Tommy, have the same weight. Find Tommy's weight.
3. On his first five Math tests, you received scores 72, 86, 92, 63, and 77. What test score he must earn on his sixth test so that his average for all six tests will be 80?
4. Sakib has four 10 km segments to his car trip. He drives his car 100 kmph for the first 10 km, 110 kmph for the second 10 km, 90 kmph for the third 10 km, 120 kmph for the fourth 10 km. What is his average speed?
5. For any two numbers, AM = 10 and GM = 8. Find out the numbers.
6. A train moves 1st 80 km at speed 75 km/h, 2nd 70 km at speed 85 km/h, 3rd 85 km at speed 66 km/h and 4th 55 km at speed 50 km/h, find the average speed throughout the journey.
7. Find an appropriate measure of central tendency with justification of the following observations:

Observations (x):185, -12, 16, -16, 0, 15.

8. An investor's return rate for the last four years is 5%, 10%, 15%, and 20%**.** Calculate the average annual return rate**.**
9. The number of employees of a multinational company of 10 countries are 49, 56, 44, 87, 29, 47, 40, 39, 46, and 50. Find the mean deviation and variance of employees.
10. A company has two sections with 40 and 65 staffs respectively. Their average weekly wages are $450 and $350. The standard deviations are 7 and 9. Which section has larger variability in wages?
11. The following data represent the number of computer centers in different localities in Dhaka. Calculate the average computers per locality.

Class interval of centers	No. of localities
10-20	8
20-30	7
30-40	5
40-50	6
50-60	4

12. Find the average speed (minute /customer) of serving customer in a customer care center. Where in different days the speeds are (x): 15, 12, 13, 19, 10.
13. The following data represent the distribution of time (minutes) needed to develop

Class interval of time	0-10	10-20	20-30	30-40	40-50	Total
No. of programs	6	8	9	8	9	40

computer programs for solving some mathematical problems.

Calculate the mean time of developing a computer program.

14. The following is the distribution of consumption of electricity (MW/locality) in different days:

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	3	12	8	7	15	26	5	3	1

Calculate harmonic mean of the distribution.

15. The distribution of number of faded signals of a day sent from different stations is as:

Class Interval of faded signals	1 – 2	2 - 3	3 – 4	4 - 5	5 - 6	Total
No. of stations (*f*)	1	3	8	6	2	20

Calculate

(a) Arithmetic, geometric and harmonic mean

(b) Median and mode of the data set

(c) Skewness and comment.

(d) Mean deviation

(e) Variance and Standard deviation

(f) Coefficient of variation

16. The analysis of the sales of 400 firms in an industry give the following distribution

Sales (in thousand Tk.)	0-50	50-100	100-150	150-200	200-250	250-300
No. of firms	8	24	55	98	120	95

Calculate

(a) Arithmetic, geometric and harmonic mean

(b) Median and mode of the data set

(c) Skewness and comment.

(d) Mean deviation

(e) Variance and Standard deviation

(f) Q_1 and Q_3, also draw a box – and – whisker plot.

17. The following frequency distribution refers to the number of hours worked per month of 50 workers of a factory. Compute median and mode of the frequency distribution.

Number of hours worked	30-55	55-80	80-105	105-130	130-155	155-180	180-205
Number of workers	3	4	6	9	12	11	5

Calculate

(a) Arithmetic, geometric and harmonic mean

(b) Median and mode of the data set

(c) Skewness and comment.

(d) Mean deviation

(e) Variance and Standard deviation

(f) Q_1 and Q_3, also draw a box – and – whisker plot.

18. What are outliers? How can we screen the data if the outliers are present in the given data?

x: 84, 10, 32, 19, 21, 29, 33, 15, 49, 2, 7, 52.

19. How do you detect and correct the information collected?

20. How do you perform with the given data having missing values?

	Col 1	Col 2	Col 3	Col 4	Col 5
0	2	5.0	3.0	6	NaN
1	9	NaN	9.0	0	7.0
2	19	17.0	NaN	9	NaN